

Summary:

Mechanical engineering senior with hands-on experience in robotics, control system design and mechanical design looking for entry level jobs in robotics and mechanical engineering

Education:

B.S. Mechanical Engineering

San Francisco State University

Spring 2026

Major Projects:

Solar and Battery Powered Boat

2022 - 2024

- Worked on the design and construction of a 50v electric boat for competition in the California solar regatta run by Sacramento's SMUD
- Lead the development of mechanical design of onboard safety systems, rudder and solar panel rotation joints, overall structure and design of hull
- Improved performance of our boat through a number of means:
 - Design of Boat structure, Stress analysis and testing, FEA using Solidworks
 - Analysis, design and testing of hydrodynamic components, BLDC motor control
 - tolerances and designing for machining, Manual machining on mill and lathe, composite carbon fiber parts
- Lead to an overall first place 2 out of 3 years attended against top engineering schools in California, with significant gains in maneuverability and efficiency

Senior capstone project: Autonomous vehicle using omni direction movement

2024 - 2025

- In a team of 3 engineers created a control system and mechanical design for an autonomous shopping cart project intended for integration with existing autonomous navigation technology
- Used engineering tools to ensure the function of the vehicle:
 - Designing electrical and mechanical system, wiring and routing using Solidworks
 - designing and fabricating custom machined parts to specific tolerances, along with FEA analysis
 - Control systems design and PID tuning, simulation through Matlab simulink
 - implementing an inverse kinematic control and control system using raspberry pi pico in C in order to meet specific design goals

Robot arm repair and control

2026

- As part of masters level courses taken at SFSU, repaired an existing robot arm developed at SFSU and implemented a joint space control method
- Diagnosed and repaired existing issues with the robot arm, involving rewiring and creating a new housing for motor drivers and joystick input methods
- Built on top of previous work by developing the kinematic and inverse kinematics model based on the product of exponentials (POE) convention
- Created a 3d assembly of the arm along with documentation in order to streamline future use by students

Skills:

CAD/Simulation: SolidWorks, Fusion 360, MATLAB, Simulink

Programming: C, Python, microcontroller programming

Controls/Robotics: PID control, robotic arm kinematics

Fabrication: CNC/manual mill and lathe, FDM 3D printing, soldering, tolerance-driven machining

Other: Excel, LaTeX, technical documentation

Professional Experience:

Customer service

Mitchell's Ice Cream, San Francisco, CA

July 2023 to present

- Assisted customers in a calm and efficient manner at one of the busiest ice cream stores in San Francisco